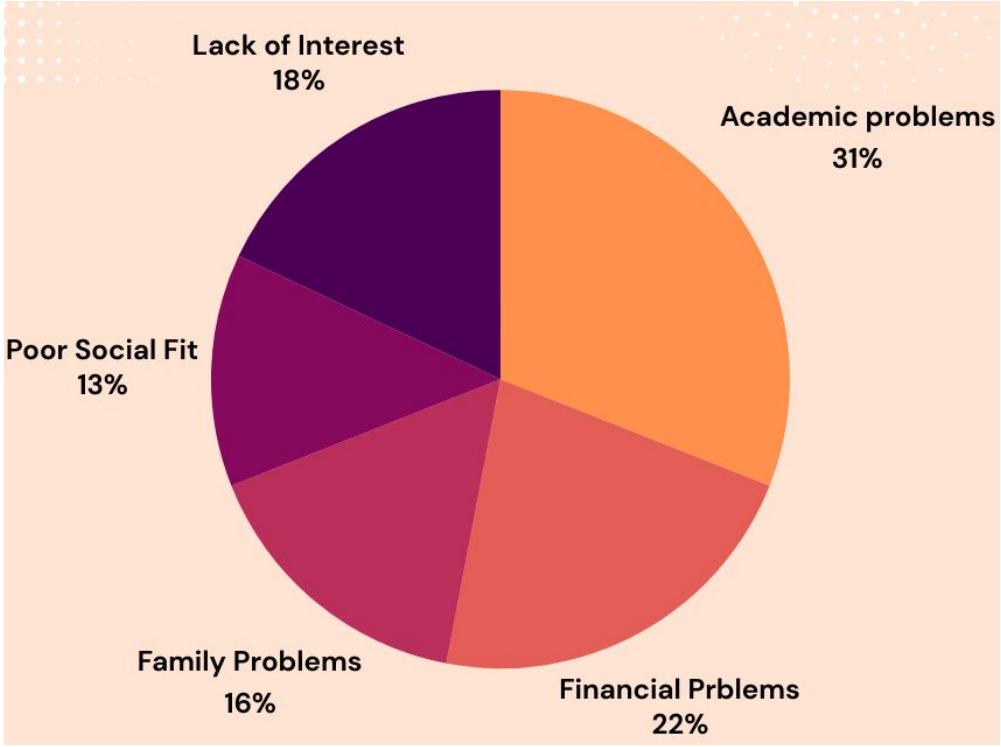
SMART INDIA HACKATHON 2024

- **Problem Statement ID –** 1661
- **Problem Statement Title-**
  Implement Software Solutions to Reduce Student Dropout Rates at Various Educational Stages
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-**
- **Team Name :** Shaswat

Innovative Software Solutions to Minimize Dropout Rates in Education

Problem Statements

- Financial Problem, Emotional Problem, Lack of Awareness.



Proposed Solution

- Utilizes predictive analytics to identify at-risk students.
- Offers academic support, emotional support, financial aid, Parents & student counseling.
- Holistic approach to reduce dropout rates by addressing both student and family challenges.

TECHNICAL APPROACH

Front end:

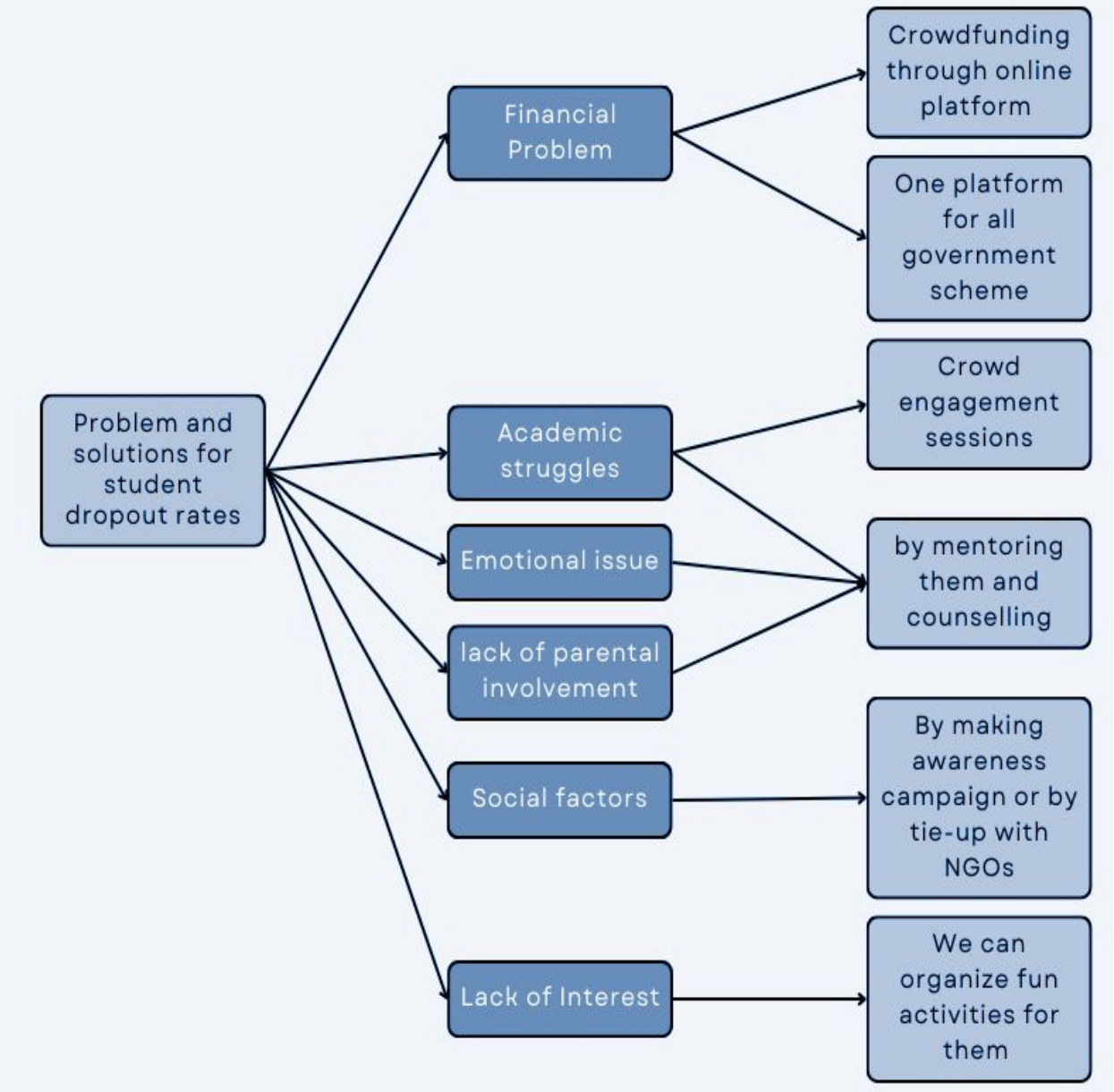
- Student registration and progress tracking
- Access to resources and counseling services
- Personalized notifications for support

Back end:

- Student data storage and dropout risk analytics
- API integration and student engagement monitoring

Hosting:

- Secure, scalable cloud platform
- Real-time data management and educational system integration



FEASIBILITY AND VIABILITY

The idea is feasible but challenging, requiring:

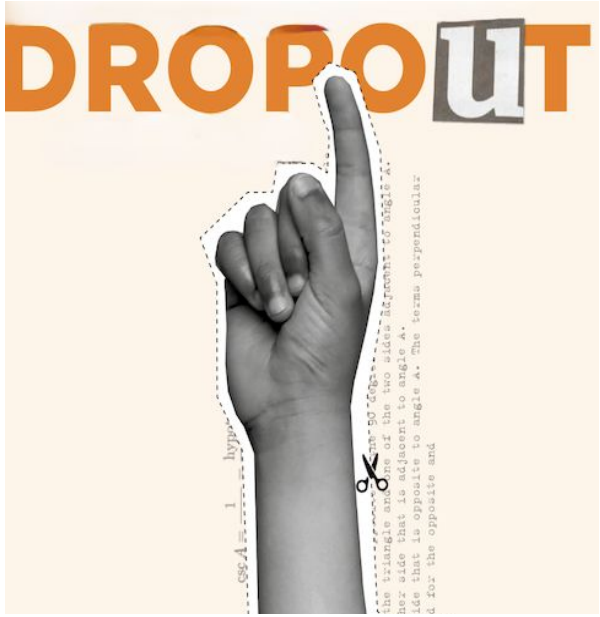
- Effective student engagement.
- Parent cooperation.
- Proper resource allocation.

Success factors include:

- Interactive counseling.
- Scalable interventions.
- Integrating technology to manage time and personalize support.

Challenges faced and their solutions:

Challenges	Solution
• Limited access to technology	Develop a mobile friendly version of the software that works offline .
• Data privacy and security	Use encryption, secure access, and follow regulations.
• . Addressing Multiple Dropout Factors:	Use a holistic approach with academic support, financial aid, counseling, and community involvement.



IMPACT AND BENEFITS

Impact:

Identifying at-risk students and providing targeted support can significantly improve retention rates.

- Improve overall academic and co-curricular performance.
- Address learning gaps and keep students engaged.

Benefits:

- Save costs by reducing dropout rates.
- Students will be aware of all **government Scholarships.**
- It will Help to **reduce Dropout ratio** of our Country.

RESEARCH AND REFERENCES

National Education Policy (NEP) 2020

https://www.education.gov.in/sites/upload_files/mhrd/files/NEP_Final_English_0.pdf

Research:

- Dropout Rates in Schools in India | Education for All in India
- Understanding India's dropout problem (indiatoday.in)
- https://timesofindia.indiatimes.com/education/news/20-6-pc-students-dropped-out-in-class-10-in-2021-22-highest-in-odisha-pradhan/articleshow/